

Project Report: DiffRect – Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation

Group 14

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Overview

This project explores the **DiffRect** framework, which applies *latent diffusion models* for semi-supervised medical image segmentation. DiffRect is designed to improve segmentation accuracy when only a small portion of the training data is labeled.

Objectives

- **Task 1:** Apply DiffRect on the ACDC Dataset.
- **Task 2:** Apply DiffRect on the Decathlon Prostate Dataset.
- **Task 3:** Apply DiffRect on the Synapse Dataset.

Methodology

DiffRect Overview

DiffRect leverages latent diffusion models to refine and propagate labels in a probabilistic latent space.

Key Components

- **Label Propagation:** Aligns sparse labels with latent representations.
- **Latent Diffusion:** Refines label information in a probabilistic latent space.

Impact

- Achieves high segmentation accuracy with limited labeled data.

Evaluation Metrics

Four key metrics were used to assess segmentation performance:

- **Dice Coefficient:** $\frac{2|A \cap B|}{|A| + |B|}$
- **HD95 (Hausdorff Distance at 95%):** Measures the maximum contour distance.
- **Jaccard Index:** $\frac{|A \cap B|}{|A \cup B|}$
- **ASD (Average Symmetric Surface Distance):** Mean symmetric boundary distance.

Task 1: ACDC Dataset

Dataset Overview

- Publicly available cardiac MRI dataset.
- 150 annotated samples with three classes: Left Ventricle, Myocardium, Right Ventricle.

Results

- **Class 1:** Dice: 0.873, Jaccard: 0.789, HD95: 3.499, ASD: 0.837
- **Class 2:** Dice: 0.881, Jaccard: 0.790, HD95: 1.473, ASD: 0.628
- **Class 3:** Dice: 0.919, Jaccard: 0.858, HD95: 2.339, ASD: 0.679
- **Average:** Dice: 0.891, Jaccard: 0.812, HD95: 2.437, ASD: 0.715

Task 2: Decathlon Prostate Dataset

Dataset Overview

- 634 prostate MRI scans from the Medical Segmentation Decathlon.
- Classes: Peripheral Zone (PZ), Transition Zone (TZ), Background.

Results

- **Class 1 (PZ):** Dice: 0.538, Jaccard: 0.388, HD95: 13.419, ASD: 4.272
- **Class 2 (TZ):** Dice: 0.766, Jaccard: 0.629, HD95: 14.563, ASD: 3.987
- **Class 3 (Background):** Dice: 0.000, Jaccard: 0.000, HD95: 50.000, ASD: 10.000
- **Average:** Dice: 0.652, Jaccard: 0.508, HD95: 13.991, ASD: 4.129

Task 3: Synapse Dataset

Dataset Overview

- The Synapse dataset includes 2211 multi-organ CT scans used for 9-class organ segmentation.
- Classes include: Background, Brain, Left/Right Hippocampus, Left/Right Amygdala, Left/Right Thalamus, and Left Cerebellum.

Challenges and Implementation

- **Data Format Conversion:** Original data was in `.npz` format and was converted to `.h5` format using custom scripts.
- **Custom Preprocessing:** The preprocessing method used in the original DiffRect paper was not publicly disclosed. Therefore, we applied standard techniques (e.g., resizing using `scipy.ndimage.interpolation.zoom`) which may not fully match the original setup.
- **Model Modification:** We adapted the model and data loader to support all 9 classes in the dataset.
- **Weight Initialization:** Initialized with pre-trained weights from the ACDC dataset: `unet_best_model.pth`.

Summary of Results

Although our preprocessing pipeline differs from the original (and possibly more optimized) setup used in DiffRect, we were able to achieve reasonably good performance:

- **Run 1 (20 iterations):** Average Dice: 0.8912, Jaccard: 0.8128, HD95: 2.437, ASD: 0.715
- **Run 2 (40 iterations):** Average Dice: 0.8918, Jaccard: 0.8093, HD95: 3.044, ASD: 0.817

Note: Due to differences in preprocessing and lack of access to full implementation details, our results may be slightly lower than those reported in the original paper. Nonetheless, we demonstrate that DiffRect can generalize well to large multi-organ segmentation tasks.

Conclusion

DiffRect provides an efficient semi-supervised solution to medical image segmentation with limited labeled data. Across diverse datasets like ACDC, Decathlon Prostate, and Synapse, the method demonstrated promising accuracy and generalizability. Future improvements can focus on increasing background class performance and enhancing multi-organ segmentation for high-resolution CT/MRI volumes.